module-01-what-is-iks - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 1 — What is IKS? (Darśana)

Sanskrit: *Bhāratīya Jñāna-Paramparā* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

The map of Indian Knowledge Systems.

> **Darśana note.** This workshop condenses Module 1's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Show side-by-side: Āryabhaṭa's value of π (3.1416, c. 499 CE) and Aryabhata-I's heliocentric hint. Ask: *"Which two of these are not in your science textbook? Why might that be?"*

Core idea · 15 min

IKS = the documented Indian knowledge traditions, from the Saraswati-Sindhu civilisation (~3000 BCE) through the colonial period.

Three useful framings:

1. **The Vedāṅga / Upaveda / Darśana taxonomy** (canonical) — six Vedic limbs, four applied sciences, six philosophical schools.
2. **The science-vs-religion problem.** Some of IKS (jyotiṣa-as-astronomy, Āyurveda-as-medicine) maps onto empirical disciplines. Some (mantras-as-ritual, kāla-as-cosmology) does not. Distinguish carefully.
3. **The historical question.** Documented dates and authors: Pāṇini (~500 BCE, grammar), Caraka (~200 BCE, medicine), Āryabhaṭa (499 CE, astronomy), Bhāskara II (1150 CE, math). These are real scholars with verifiable corpora.

Honest framing: IKS is neither uniformly ancient nor uniformly scientific. It is a documented intellectual tradition with reform, debate, and dates.

Demonstration · 10 min

Project Āryabhaṭa's sine table (rough chart, 24 entries, c. 499 CE) and Madhava's infinite series for π (~1400 CE, 250 years before Leibniz).

Show the Mādhava formula:

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$$

Discussion: *"This is the Leibniz series, attested in Kerala 250 years before Leibniz. Why isn't this in your maths textbook? What does its absence tell us about the politics of mathematical history?"*

Source moment · 5 min

Read aloud Āryabhaṭīya, *Gaṇitapāda* 1: *"brahmā kṛtaṁ paramaṁ rāhuṁ ca śaśinaṁ ravim…"* (the invocation), followed by Āryabhaṭa's heliocentric hint in *Gola-pāda* — *"the apparent westward motion of stars at the equator is due to the earth itself moving eastward."* Read alongside the matching paragraph from a 21st-c. NCERT physics textbook on Earth's rotation.

Hands-on activity · 15 min

Source-criticism exercise. Distribute 4 short paragraphs (cut from popular media articles claiming "ancient Indian science discovered X"). Pairs evaluate each:

1. Is the claim verifiable? (Author? Date? Manuscript?)
2. What is the manuscript evidence?
3. Is the modern parallel accurate, exaggerated, or invented?

Each pair classifies the 4 claims as: *real (documented)*, *partially exaggerated*, *anachronistic*, *fabricated*. Class discusses.

Reflection + take-home · 10 min

"Distinguish three things: (1) what's ancient AND scientifically valid, (2) what's ancient AND historically interesting BUT no longer scientifically used, (3) what's claimed as ancient BUT is actually a modern reconstruction. Give one example of each from your prior knowledge."

Go deeper into Adhyayana

The 10 days you're skipping if you only do this workshop:

- **Day 01** — Why IKS now? The mapping problem
- **Day 02** — Vedas and Vedāṅgas: the six supporting sciences
- **Day 03** — Upavedas: applied disciplines
- **Day 04** — Itihāsa and Purāṇas: history-as-story
- **Day 05** — Six Darśanas: the philosophical schools
- **Day 06** — Named figures: Pāṇini to Āryabhaṭa
- **Day 07** — Source discipline: what we can verify
- **Day 08** — IKS meets modern science: bridges and tensions
- **Day 09** — NEP 2020 and IKS integration
- **Day 10** — Synthesis project: your map of IKS

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.